

TYPES OF CHEMICAL REACTIONS

Student Learning Pack

1. Metadata

Topic	Types of Chemical Reactions
Level	Class X
Lesson Length	60 minutes

2. Learning Objectives

- LO1: Identify and describe combination reactions, noting that they are often exothermic, and write balanced equations for them.
- LO2: Distinguish decomposition reactions from other types and construct balanced equations for given examples.
- LO3: Explain single displacement, double displacement, and precipitation reactions with textbook examples.
- LO4: Identify oxidised and reduced substances in redox reactions and describe the underlying electron transfer.
- LO5: Compare exothermic and endothermic reactions, linking the concepts to the reaction types studied.

3. Prerequisite Knowledge

- Atoms are conserved during chemical reactions; no atoms are created or destroyed.
- A chemical equation must be balanced to reflect conservation of atoms.
- Basic symbols for states of matter (s, l, g, aq).
- Understanding of heat energy as a form of energy transferred during reactions.

4. Key Terms

Term	Definition
Combination reaction	Two or more substances combine to form a single new substance.
Decomposition reaction	A single substance breaks down into two or more different substances.
Single displacement reaction	An element displaces another element from its compound.
Double displacement reaction	Two compounds exchange ions to form two new compounds.
Precipitation reaction	A type of double displacement reaction that produces an insoluble solid (precipitate).
Oxidation	Gain of oxygen or loss of hydrogen in a substance during a reaction.
Reduction	Loss of oxygen or gain of hydrogen in a substance during a reaction.
Exothermic reaction	A reaction that releases heat to the surroundings.
Endothermic reaction	A reaction that absorbs heat from the surroundings.

Term	Definition
Balanced chemical equation	An equation in which the number of atoms of each element is the same on both sides.

5. Key Points

- When two or more substances combine to form a single product, the reaction is a combination reaction.
- Many combination reactions release heat, making them exothermic.
- A balanced equation has the same number of atoms of each element on both sides, reflecting conservation of mass.
- In redox reactions one reactant is oxidised and the other is reduced.
- When ions exchange between two compounds, a solid precipitate may form.
- The textbook excerpt does not provide specific decomposition examples, but the type involves a single compound breaking into two or more simpler substances.
- The provided pages do not detail single displacement reactions; they involve an element replacing another in a compound.

6. Real-World Applications

Energy from Respiration

Respiration in the human body is an exothermic process where glucose reacts with oxygen to release energy, carbon dioxide, and water.

Think about: Why does the body feel warm after a meal, based on the chemical nature of respiration?

Suggested time: 4 minutes

Combustion of Natural Gas for Home Heating

Burning natural gas (CH_4) in furnaces provides heat for homes; the reaction is exothermic, converting chemical energy to thermal energy.

Think about: How does the exothermic nature of the CH_4 combustion reaction make it useful for heating?

Suggested time: 3 minutes

7. Core Notes

Combination Reactions

- Definition: Two or more elements or compounds combine to give one product.
 - Example (burning of coal): $\text{C(s)} + \text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)}$ (1.15)
 - Example (formation of water): $2\text{H}_2\text{(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{H}_2\text{O(l)}$ (1.16)
- These equations show the reactants on the left and the single product on the right, matching the textbook description.

Exothermic Nature of Combination Reactions

- Exothermic reaction: heat is evolved, the mixture feels warm (Activity 1.4 observation).
- Example (burning natural gas): $\text{CH}_4\text{(g)} + 2\text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$ (1.17)

- Respiration is also exothermic: $\text{glucose} + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$ releases energy for the body. The textbook links the release of heat to the term “exothermic chemical reactions.”

Balancing Chemical Equations

- Principle: atoms are neither created nor destroyed.
 - Balanced example from the book: $\text{Zn} + \text{H}_2\text{SO}_4 \rightarrow \text{ZnSO}_4 + \text{H}_2$ (1.3). Atom count shows 1 Zn, 2 H, 1 S, 4 O on each side.
 - Unbalanced example to practice: $\text{Fe} + \text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + \text{H}_2$ (1.4). Students must adjust coefficients so that Fe, H, and O atoms are equal on both sides.
- Balancing ensures the law of conservation of mass is satisfied.

Oxidation-Reduction (Redox) Reactions

- Definition: simultaneous oxidation (loss of electrons) and reduction (gain of electrons).
 - Example (metal oxide reduction): $\text{ZnO} + \text{C} \rightarrow \text{Zn} + \text{CO}$ (1.31). Carbon is oxidised ($\text{C} \rightarrow \text{CO}$) and zinc oxide is reduced ($\text{ZnO} \rightarrow \text{Zn}$).
 - Example (acid-metal oxide reaction): $\text{MnO} + 2\text{HCl} \rightarrow \text{MnCl}_2 + \text{H}_2\text{O} + \text{Cl}_2$ (balanced from 1.32). MnO is reduced to Mn^{2+} , while chloride ions are oxidised to Cl_2 gas.
- These illustrate electron transfer between reactants.

Double Displacement and Precipitation Reactions

- Definition: exchange of ions leading to new compounds; if an insoluble product forms, it appears as a precipitate.
 - Classroom observation (Activity 1.2): mixing lead(II) nitrate and potassium iodide produced a white precipitate of lead(II) iodide.
 - The other product, sodium chloride, stayed dissolved.
- The reaction exemplifies a double-displacement (metathesis) reaction that yields a precipitate.

Decomposition Reactions

- General idea (from curriculum): A decomposition reaction splits one reactant into multiple products, often requiring heat or electricity.
- No explicit equation is given in the supplied pages, so students should refer to other textbook sections for concrete examples.

Single Displacement Reactions

- General concept (curriculum): An element (often a metal) displaces another element from a compound, forming a new compound and a free element.
- No specific reaction is presented in the source material, so learners should consult additional textbook sections for examples.

8. Activities

Write and Balance Combination Equations (12 min)

Students choose two substances (e.g., carbon and oxygen, hydrogen and oxygen) and write the correct chemical equation. Then they balance the equation, checking atom counts on both sides.

Identify Oxidised and Reduced Species (13 min)

Using the redox examples $\text{ZnO} + \text{C} \rightarrow \text{Zn} + \text{CO}$ and $\text{MnO} + 2\text{HCl} \rightarrow \text{MnCl}_2 + \text{H}_2\text{O} + \text{Cl}_2$, students label which reactant is oxidised and which is reduced, and write the half-reactions.

9. Questions

Q1. What is a combination reaction? Give an example from the textbook.

Q2. Write the balanced chemical equation for the decomposition of zinc carbonate as presented in the source.

Q3. When potassium bromide reacts with barium iodide, what type of reaction occurs and what are the products?

Q4. In the reaction $4\text{Na(s)} + \text{O}_2\text{(g)} \rightarrow 2\text{Na}_2\text{O(s)}$, which substance is oxidised and which is reduced?

Q5. Explain why the reaction of calcium oxide with water is described as exothermic.